ComicCraft – AI-Powered Story to Comic Generator

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Project Name: ComicCraft
Project Type: AI-Powered Web Application
Project Domain: Artificial Intelligence, Natural Language Processing, and Visual Storytelling
Documentation Folder: 4_Project_Planning
Documentation File: README.md
Primary Workflow: Story Input → Story Processing → Comic Panel Generation → Final Comic

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1. Project Overview

ComicCraft is an AI-powered application designed to convert written stories into visually engaging comic panels. The application accepts a story from the user, processes its narrative, divides it into meaningful scenes, generates or retrieves illustrations, integrates dialogue and captions, and assembles the panels into a final comic.

The purpose of this project-planning document is to define the activities, objectives, scope, schedule, responsibilities, resources, risks, and expected deliverables required to develop ComicCraft.

Project planning helps ensure that the development process is organized, measurable, and aligned with the project's objectives. It also helps the development team identify dependencies, manage resources, monitor progress, and test the application systematically.

2. Problem Statement

Creating comics manually requires several activities, including story writing, character design, scene planning, illustration, dialogue placement, and page layout.

These activities can consume significant time and require artistic or technical skills. Users who have creative ideas but lack drawing experience may find it difficult to turn their stories into complete comics.

ComicCraft addresses this challenge by proposing an automated workflow that transforms written stories into structured comic panels and combines them into a final visual narrative.

The project plan focuses on developing this workflow in manageable stages, from gathering requirements to testing and delivering the application.

3. Project Objectives

The main objectives of ComicCraft are:

- To develop a user-friendly interface for entering written stories.
- To validate story input before processing.
- To analyze story content and identify important characters, events, and settings.
- To divide the story into meaningful and ordered scenes.
- To generate structured descriptions for individual comic panels.
- To generate or retrieve illustrations for the panels using the selected visual-generation component.
- To integrate dialogue, captions, and scene descriptions.
- To arrange the panels into a readable comic layout.
- To provide a final comic preview.
- To support comic editing, saving, or exporting where these features are implemented.

- To test the complete workflow and improve reliability.
- To maintain project documentation and source code using GitHub.

4. Detailed Project Summary

ComicCraft follows a planned development process in which a written story is progressively transformed into a complete comic.

The project begins with requirement analysis and interface planning. The team then develops the story-input functionality, story-processing component, scene-segmentation logic, panel-generation workflow, visual-generation integration, and comic-layout functionality.

During execution, each component is developed and tested separately before the components are integrated into the complete application.

4.1 Story Input Planning

The application will provide an interface where users can enter or paste their stories.

Planning activities include defining input requirements, setting supported story-length limits, designing input-validation rules, and deciding how invalid or empty submissions will be handled.

Planned input: Written story text.

Expected result: A validated story ready for processing.

4.2 Story Processing Planning

The development team will select an appropriate method for analyzing the submitted story.

Depending on the chosen implementation, this component may use Natural Language Processing (NLP), rule-based processing, or a language-model API.

Planning activities include identifying the story-processing approach, defining the required output structure, and determining how characters, locations, events, and dialogue will be represented.

Planned output: Structured story information.

4.3 Scene Segmentation Planning

The story will be divided into scenes that represent important events in the narrative.

The team will determine how scenes are identified, how their order is maintained, and how each scene is represented in the application's internal data structure.

Planned output: An ordered list of scenes.

4.4 Comic Panel Generation Planning

The panel-generation component will transform each scene into a structured panel description.

The plan includes defining panel metadata, visual descriptions, character actions, facial expressions, background details, dialogue, and captions.

Planned output: Panel descriptions ready for visual generation and layout.

4.5 Visual Generation Planning

The project will use an appropriate image-generation service, model, or illustration source if visual generation is part of the selected implementation.

The team will plan prompt preparation, image association, error handling, retry behavior, and any supported fallback for failed image requests.

Planned output: Illustrations associated with individual panels.

4.6 Dialogue and Caption Planning

The application will associate dialogue and narration with the appropriate panels.

The plan includes maintaining text readability, preserving the meaning of the original story, and preventing dialogue from being assigned to the wrong scene.

Planned output: Panels containing illustrations and associated text.

4.7 Comic Assembly Planning

The assembly component will arrange the panels into a consistent visual sequence.

The team will define panel ordering, spacing, borders, text placement, page dimensions, and the final preview format.

Planned output: An assembled comic layout.

4.8 Final Comic Delivery